

Alikasim Budhwani

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SKILLS

- **Programming Languages:** Python, C++, MATLAB, Bash
- **Toolboxes/Environments:** ROS/ROS2, MoveIt, PyTorch, OpenCV, MuJoCo, Isaac Sim, CAN-FD, Modbus/PLC, Docker, Linux

EXPERIENCE

- **Autonomique** Mar 2026 - Present
Robotics Engineer Menlo Park, USA
 - Train and deploy learned manipulation policies (ACT, Diffusion Policy) for a bushing press task on a dual-arm humanoid in production at a Tier-1 automotive supplier, achieving 96% placement and insertion success with sub-3 s insertion times and onboard Jetson AGX Orin inference at 50 Hz.
 - Architect a layered autonomy stack separating the production workflow from per-state strategies and hardware adapters, making it robot-agnostic and portable to any humanoid platform.
 - Implement coordinated dual-arm picking and placing using 6-DoF pose tracking and visual servoing, integrating MoveIt motion planning, hand-eye calibration, grippers, PLC press I/O, and mobile-base navigation into a single production cycle.
 - Develop VR (Meta Quest/WebXR) and SpaceMouse teleoperation for demonstration collection, streaming low-latency CAN-FD commands to sustain 100 Hz continuous control, up from 10 Hz.
- **BHF Robotics** Sep 2023 - Dec 2024
Robotics Engineer Markham, Canada
 - Led a team of 4 interns in developing ROS-based autonomy stack for Agrobot mobile platform which conducts targeted weed elimination via high-voltage electricity.
 - Optimized weed elimination sequencing using Ant Colony Optimization (ACO), achieving a high-speed cycle time of 300 weeds/min.
 - Developed the perception module for precise weed detection and elimination site selection, minimizing crop damage while achieving 90% accuracy.
- **Ansys** Jan 2023 - Apr 2023
SDE Intern Waterloo, Canada
 - Implemented isoparametric mapping in C++-based finite element solver, reducing computation time by 10% for 10,000+ node models with complex curved geometries.
 - Refactored C++ element classes in coupling framework using templates, eliminating 15+ redundant classes and improving readability.
- **Promation** Sep 2021 - Dec 2021
Perception Intern Oakville, Canada
 - Developed an in-situ defect detection system for 3D metal printing, obtaining 75% detection accuracy.
 - Implemented a custom hand-eye calibration algorithm for 2D laser scanners, reaching 0.5mm accuracy during 3D object reconstruction.

RESEARCH

- **Latent Safety Filters for Contact-Rich Manipulation** Sep 2025 - Present
MASc thesis, University of Waterloo (Prof. Soo Jeon).
 - Developing a label-free safety monitor that flags unsafe policy actions before execution, using trajectory divergence under small action perturbations as a proxy for imminent failure.
 - Scores divergence by rolling perturbed actions through a DINOv2-based latent world model; validated on 70 diffusion-policy rollouts against MuJoCo ground truth (96.9% accuracy, 0.62 F1).
- **WATonomous** Feb 2022 - Apr 2023
University of Waterloo's Autonomous Vehicle Team.
 - Led the planning and controls team in implementing a real-time MPC module in ROS2.
 - Developed a spatio-temporal action recognition model to predict pedestrian behavior, improving video-level mAP@0.2IOU by 2.2 points over the existing baseline.

EDUCATION

- **University of Waterloo** Jan 2025 - Apr 2027
MASc in Mechanical Engineering (GPA: 4.00/4.00) Waterloo, Canada
 - **Coursework** including Nonlinear Control, Stochastic Control, and Vehicle Dynamics.
- **University of Waterloo** Sep 2019 - Aug 2024
BASc in Mechanical Engineering (GPA: 3.84/4.00) Waterloo, Canada
 - **Coursework** including Robot Manipulation, Mobile Robotics, and Reinforcement Learning.